

PANAMA PACIFICO, Panama

WMO: 787955

Lat: 8.917N

Lon: 79.600W

Elev: 13

StdP: 101.17

Time Zone: -5.00 (W05)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	22.2	22.9	18.0	13.0	31.5	18.9	13.7	30.4	10.1	27.5	9.3	27.2	1.3	10	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.8	35.0	25.4	34.1	25.3	33.8	25.2	27.7	31.7	27.2	31.2	26.9	30.8	5.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.2	30.5	26.1	21.6	29.9	25.9	21.3	29.7	88.4	32.2	86.4	31.7	84.9	31.2	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	20.7	36.6	0.7	0.7	20.2	37.1	19.7	37.5	19.3	37.9	18.8	38.4		
(5)			19.5	29.4	0.6	0.8	19.0	30.0	18.7	30.4	18.3	30.8	17.9	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	28.2	28.8	29.2	29.3	28.6	28.0	28.0	27.7	27.5	27.1	27.2	27.6
	DBStd	1.30	1.00	0.96	1.08	1.14	1.27	1.20	1.28	1.11	1.12	0.98	0.90	0.96
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6607	566	525	595	579	576	541	559	549	524	531	516	547
	CDD18.3	3566	307	292	337	329	318	291	301	291	274	272	266	288
	CDH23.3	35396	3017	3036	3717	3687	3322	2952	3011	2804	2543	2355	2283	2671
	CDH26.7	12436	1152	1305	1689	1623	1179	944	943	819	714	616	573	877
Wind	WSAvg	3.0	3.6	4.2	4.2	3.6	2.8	2.3	2.7	2.5	2.4	2.5	2.4	2.8
Precipitation	PrecAvg	1799	51	18	11	86	189	205	164	200	191	284	274	127
	PrecMax	2586	263	97	51	216	340	431	289	328	330	617	660	274
	PrecMin	1288	0	0	0	7	114	81	68	90	86	135	127	20
	PrecStd	358	72	24	14	69	68	92	64	68	62	125	117	72
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	35.1	35.2	36.1	35.0	33.8	33.8	33.1	33.1	32.2	32.1	33.2
		MCWB	25.5	25.3	25.1	24.9	26.7	27.5	27.1	26.9	26.7	26.6	26.0	26.1
	2%	DB	33.2	34.1	34.9	35.0	33.9	32.2	32.8	32.1	32.0	31.2	31.2	32.2
		MCWB	25.0	24.6	25.0	25.4	26.7	26.8	26.8	26.6	26.5	26.4	26.0	25.4
	5%	DB	32.8	33.2	34.1	34.1	32.8	31.9	31.9	31.1	31.0	30.8	30.8	31.9
		MCWB	24.9	24.3	24.7	25.3	26.4	26.6	26.5	26.6	26.4	26.3	25.9	25.2
	10%	DB	31.9	32.9	33.2	33.1	31.8	30.9	30.9	30.2	30.1	29.9	29.9	30.9
		MCWB	24.6	24.3	24.5	25.2	26.2	26.5	26.5	26.1	26.0	25.8	25.6	25.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	26.8	27.2	28.0	28.2	28.1	28.0	27.6	27.4	27.3	27.1	27.4
		MCDB	31.9	33.1	33.5	32.6	32.7	31.8	31.8	31.6	31.3	30.6	30.6	31.6
	2%	WB	26.6	26.2	26.3	26.8	27.2	27.4	27.3	27.0	26.8	26.7	26.5	26.5
		MCDB	31.2	32.1	32.7	31.9	31.4	31.0	31.3	30.8	30.4	30.0	30.0	30.7
	5%	WB	26.0	25.6	25.7	26.2	26.7	26.9	26.7	26.6	26.5	26.3	26.1	26.0
		MCDB	30.3	31.1	31.8	31.4	30.6	30.3	30.5	30.1	29.9	29.5	29.5	30.2
	10%	WB	25.4	25.0	25.2	25.9	26.5	26.5	26.4	26.3	26.1	26.0	25.7	25.6
		MCDB	29.6	30.1	30.9	30.9	30.0	29.7	29.8	29.5	29.2	29.1	28.8	29.4
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.9	9.4	8.8	6.9	6.3	6.3	6.2	6.2	6.2	6.0	7.1
		MCDBR	8.8	9.6	9.9	9.7	8.2	7.3	7.3	7.1	7.1	7.2	6.9	8.1
		MCWBR	3.0	2.9	3.0	3.1	3.1	3.3	3.4	3.4	3.4	3.5	3.2	3.3
	5% WB	MCDBR	7.6	8.5	8.9	8.4	7.1	6.6	6.6	6.5	6.5	6.6	6.2	7.2
		MCWBR	3.3	3.2	3.3	3.4	3.2	3.3	3.4	3.3	3.3	3.4	3.2	3.3
Clear-Sky Solar Irradiance	taub	0.422	0.451	0.465	0.479	0.469	0.445	0.456	0.459	0.429	0.414	0.433	0.425	
	taud	2.413	2.319	2.283	2.250	2.315	2.386	2.360	2.347	2.424	2.493	2.427	2.434	
	Ebn at Noon	889	873	862	838	830	840	834	844	880	896	872	878	
	Edh at Noon	118	133	140	143	131	120	124	128	119	110	116	114	
All-Sky Solar Radiation	RadAvg	5.21	5.64	5.68	5.50	4.60	4.20	4.16	4.29	4.41	4.31	4.02	4.54	
	RadStd	0.24	0.26	0.29	0.24	0.34	0.27	0.26	0.23	0.25	0.21	0.32	0.46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	-0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page